

Experiment 6

Module Name: Automation with Selenium

Experiment Title: Windows, Tabs, and Iframes

Problem Statement: Interact with elements inside an iframe, open a new tab, switch to it, and return to the main layout.

Objective: Master context switching using `switch_to.frame()` and `window_handles`.

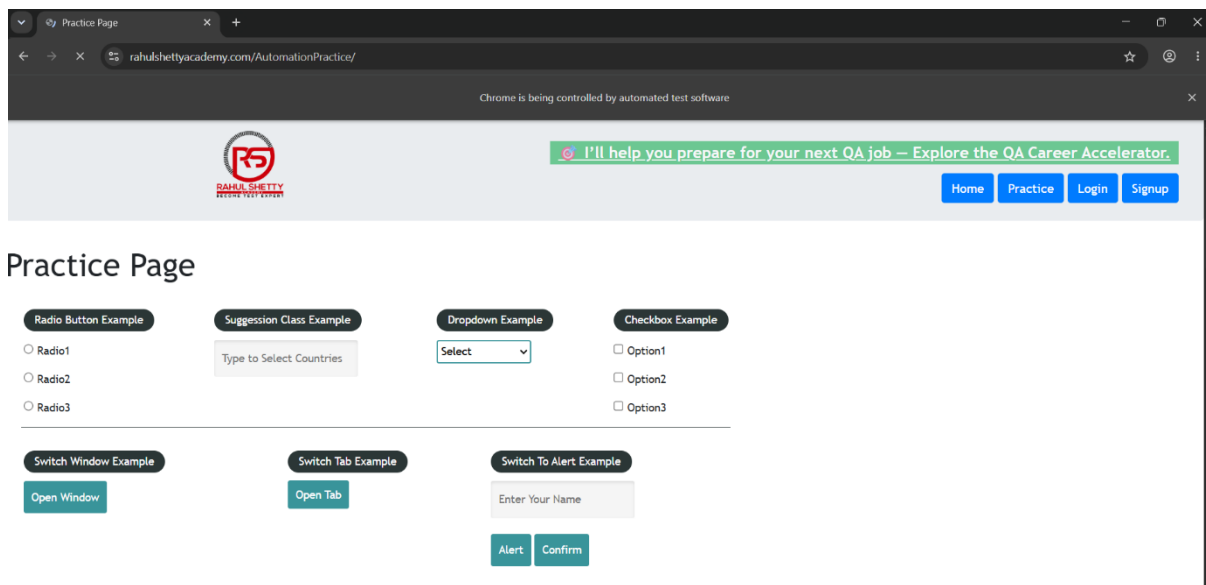
Tools, software, and concepts used: Python, Selenium, Iframe handling, Window Handlers.

Source code or implementation steps:

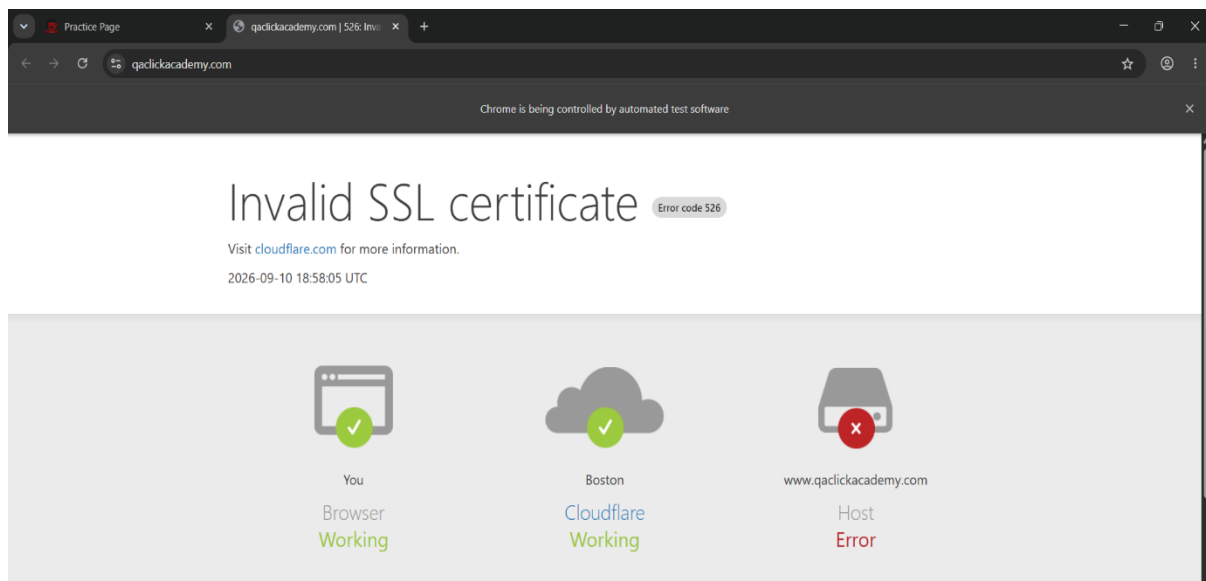
```
import time
from selenium import webdriver
from selenium.webdriver.common.by import By
driver = webdriver.Chrome()
driver.maximize_window()

try:
    driver.get("https://rahulshettyacademy.com/AutomationPractice/")
    main_window = driver.current_window_handle
    # 1. Switch to Iframe and interact
    driver.switch_to.frame("courses-iframe")
    iframe_text = driver.find_element(By.XPATH, "//*[div[@class='header-upper']//li[contains(text(), 'contact')]]").text
    print(f"Extracted from Iframe: {iframe_text}")
    # Switch back to default content
    driver.switch_to.default_content()
    # 2. Open new tab and switch
    driver.find_element(By.ID, "opentab").click()
    time.sleep(2) # Allow tab to load
    all_windows = driver.window_handles
    for window in all_windows:
        if window != main_window:
            driver.switch_to.window(window)
            break
    print(f"New Tab Title: {driver.title}")
    driver.close() # Close the child tab
    # Switch back to main window
    driver.switch_to.window(main_window)
    print(f"Returned to Main Window Title: {driver.title}")
finally:
    driver.quit()
```

Output with clearly labelled screenshots:



Screenshot 1: Main page with embedded iframe



Screenshot 2: New browser tab opened by the script

Console Output: Extracted from Iframe: contact@rahulshettyacademy.com

New Tab Title: QAClick Academy - A Testing Academy to Learn, Earn and Shine

Returned to Main Window Title: Practice Page

Brief result: Context successfully shifted between frames and OS-level browser tabs.

Observation: Selenium operates in one context at a time. Elements inside an iframe or new tab are completely invisible to the driver until an explicit switch_to command is executed.

Conclusion: Managing window_handles and default_content() is crucial for complex web applications.